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Guide Name: Dr. P Jyothi

TITLE: Real-Time Emergency Vehicle Detection and Dynamic Traffic Control using YOLOv8

Project Team No: 03

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Abstract:

Rapid urbanization and the steady increase in traffic congestion have become major challenges in providing timely emergency medical services in modern cities. Ambulances often experience significant delays at signalized intersections because conventional traffic signals operate on fixed-time schedules and lack the ability to coordinate with emergency vehicles in real time. These delays increase emergency response time, which can directly affect patient outcomes and reduce the chances of saving lives during critical situations. Existing traffic management systems primarily focus on maintaining general traffic flow and offer limited support for prioritizing emergency vehicles.

To address this problem, this project proposes an AI-driven Adaptive Traffic Control System that creates intelligent green corridors for ambulances by integrating GPS-based route prediction, edge-based computer vision, adaptive traffic signal control, and lightweight communication technologies. The system uses GPS data to predict the ambulance's route and notify upcoming intersections before its arrival, allowing sufficient time for traffic signal preparation. A custom-trained YOLOv8 model deployed on edge devices processes live traffic camera feeds to detect ambulances and estimate traffic conditions in real time. The system calculates vehicle queue length and Time-To-Clear (TTC) to determine the optimal signal preemption timing, ensuring efficient traffic clearance while minimizing disruption to regular traffic flow. MQTT-based communication enables coordination among multiple intersections to provide a continuous green corridor throughout the ambulance's journey.

The proposed solution is designed to be scalable, reliable, and cost-effective by leveraging existing traffic camera infrastructure along with low-cost edge devices, making it suitable for practical deployment in urban environments. Its performance is evaluated using metrics such as detection accuracy, response latency, queue clearance efficiency, and reduction in ambulance travel time to assess its effectiveness in improving emergency response.

Keywords:

Adaptive Traffic Control System, Emergency Vehicle Priority, YOLOv8, Edge Computing, GPS-Based Route Prediction, Traffic Signal Preemption, MQTT Communication, Intelligent Transportation Systems (ITS).

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